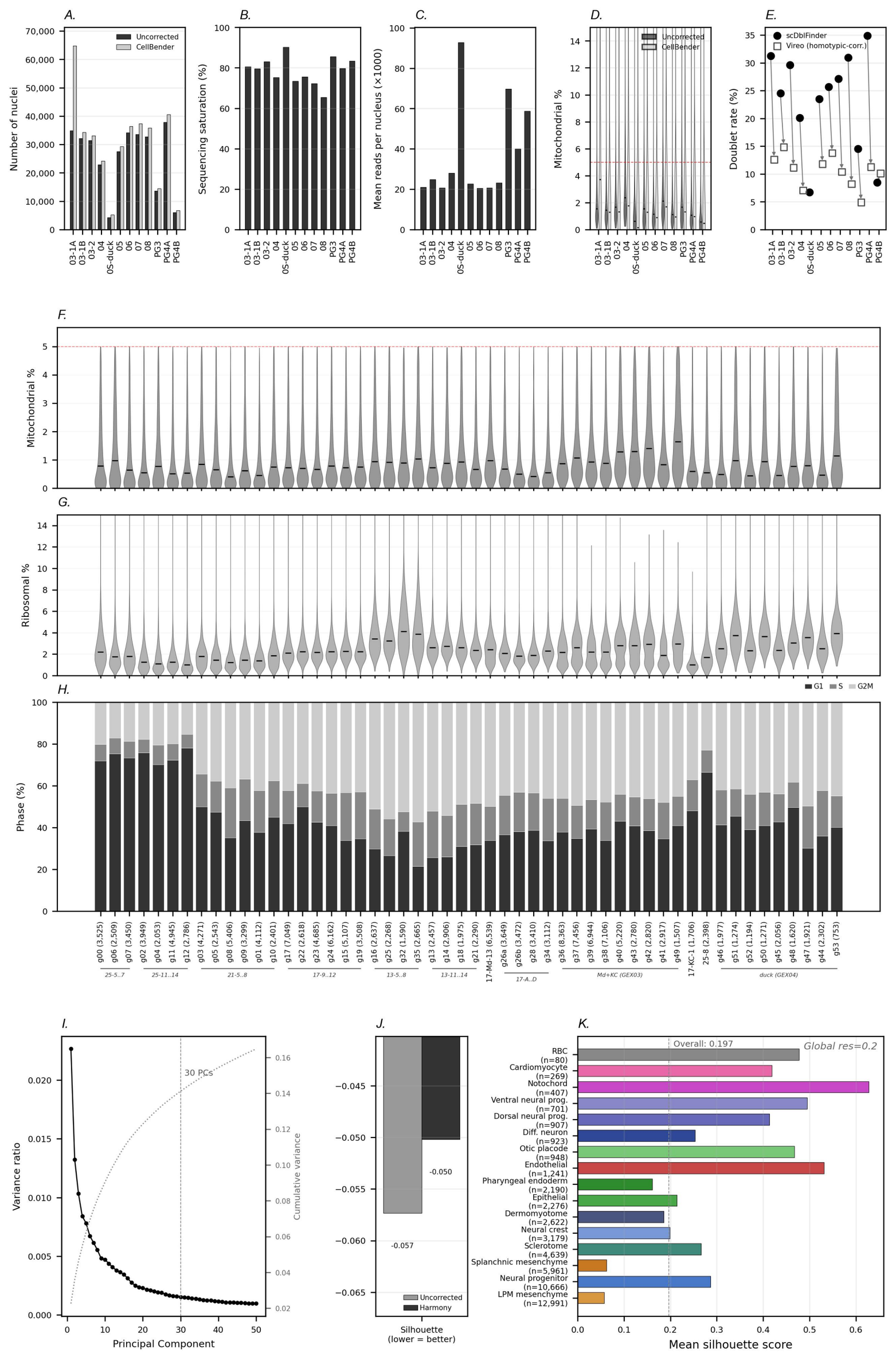
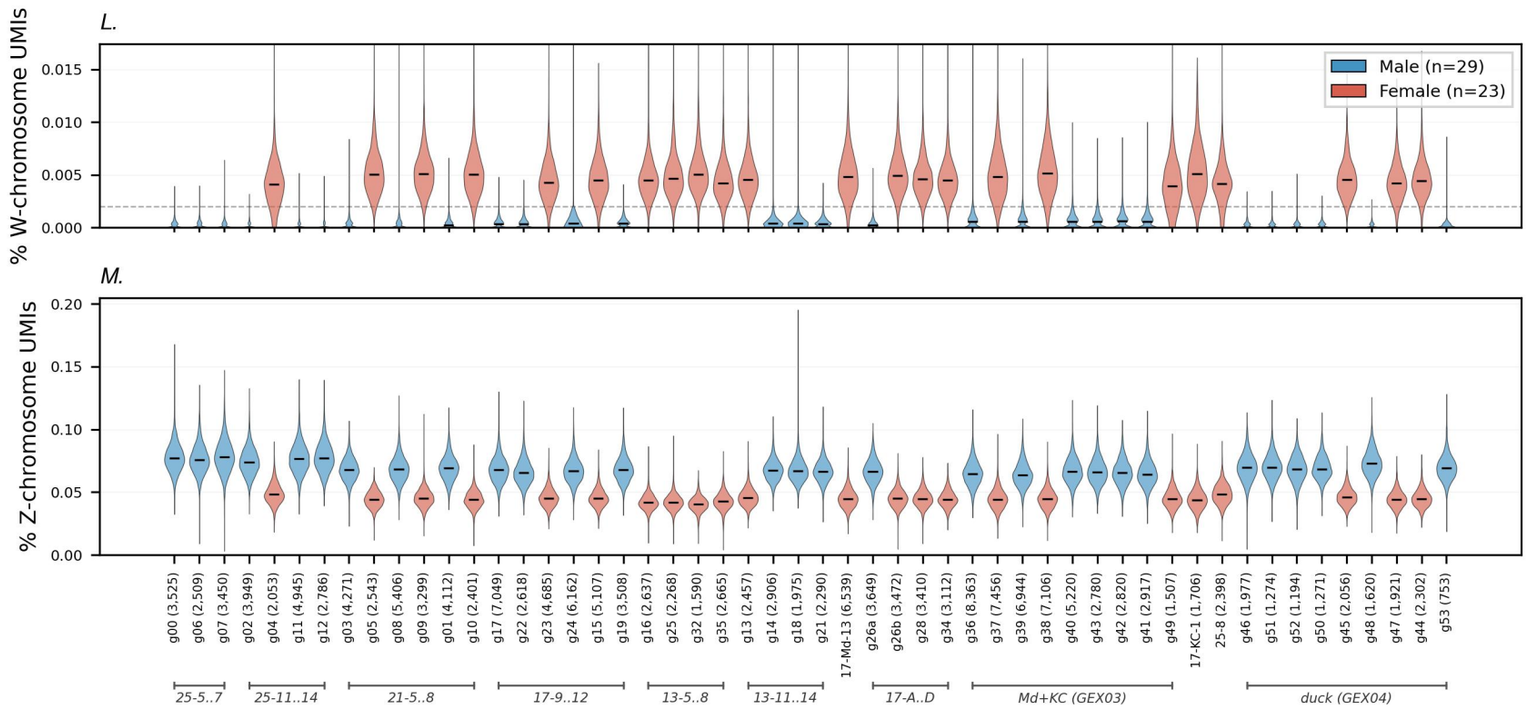
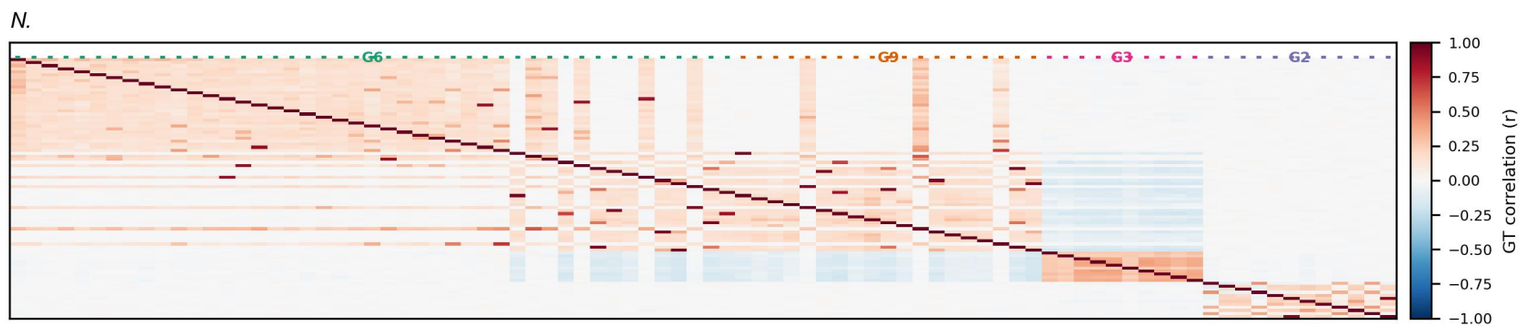
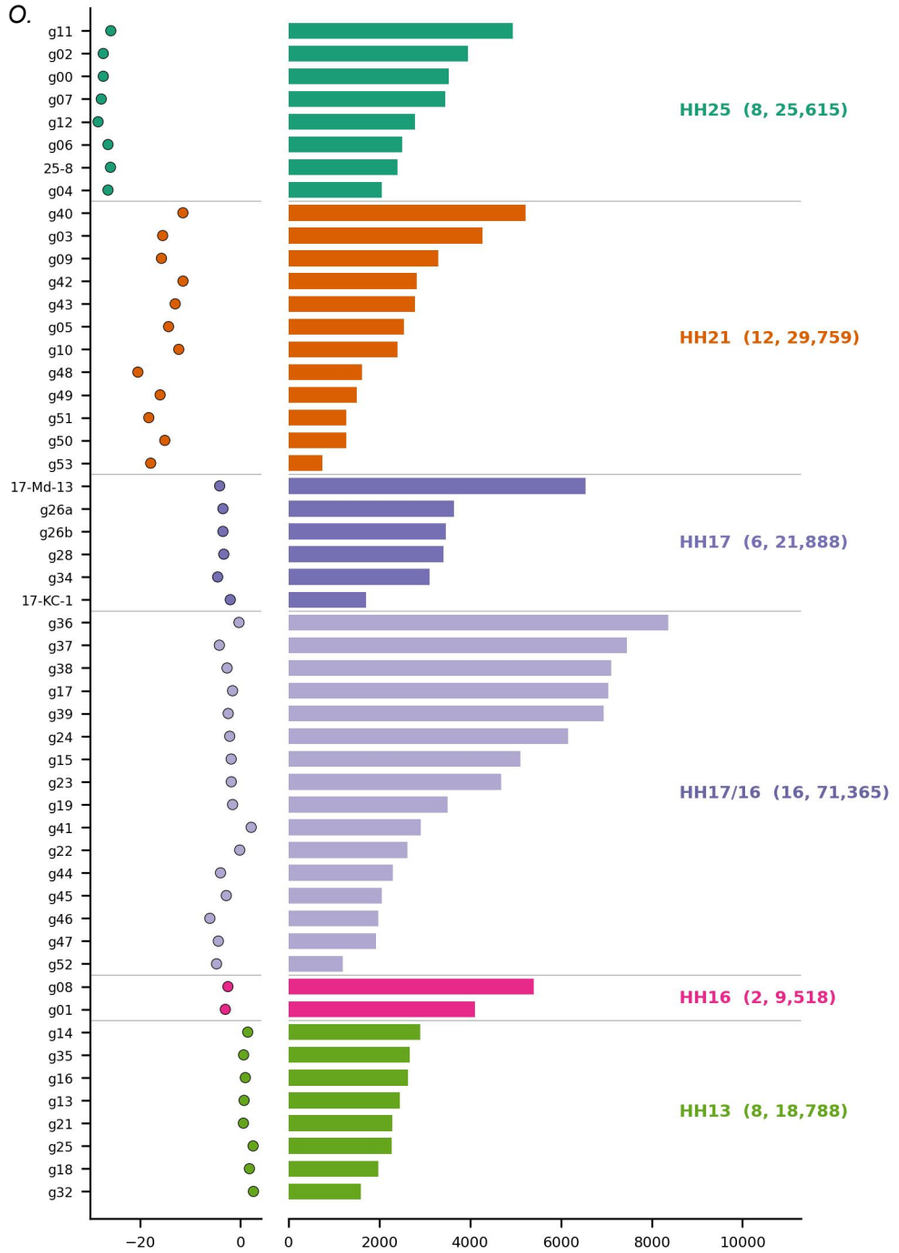
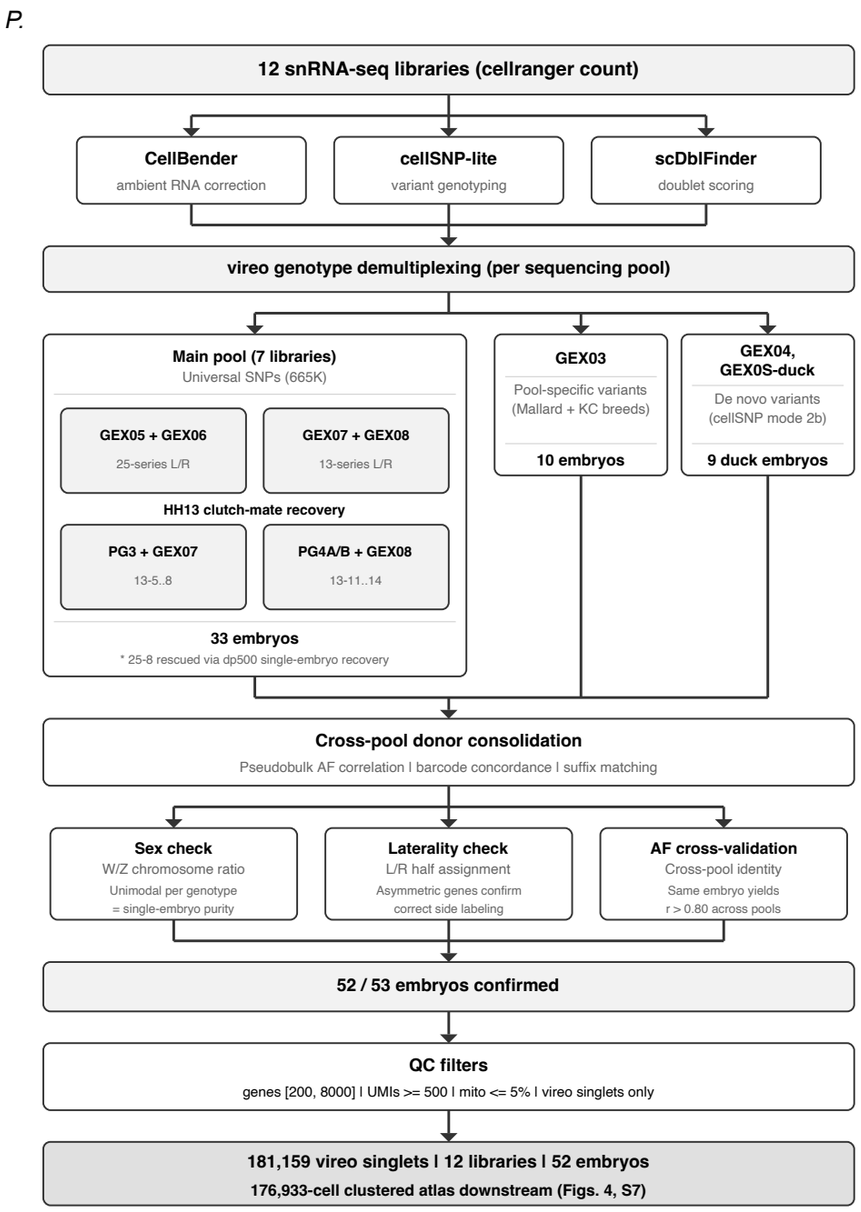
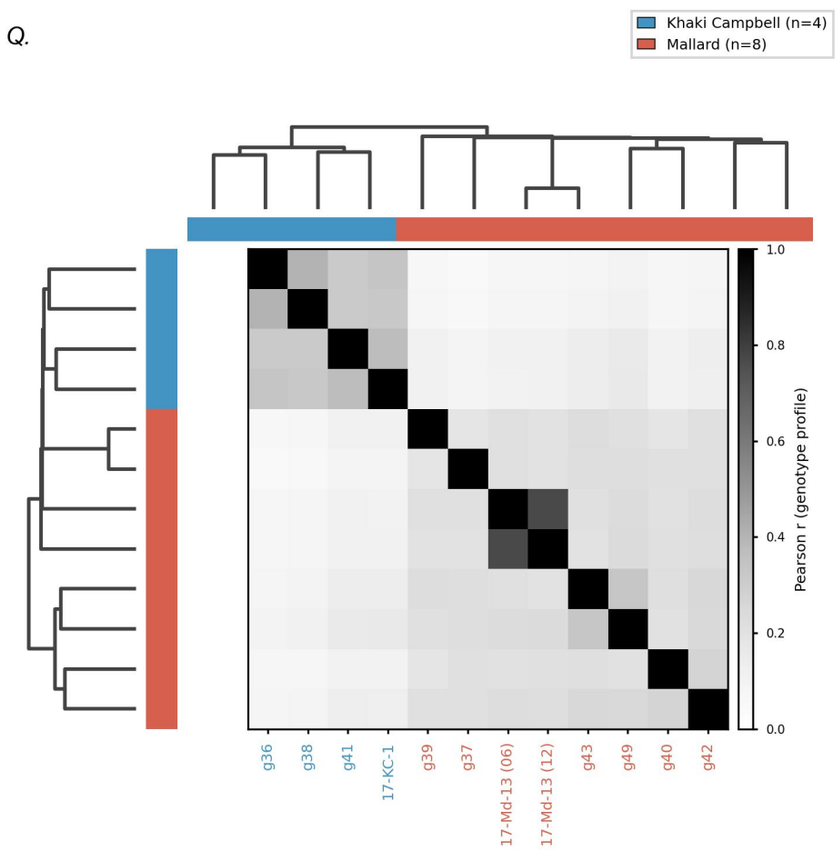

A.
Number of nuclei
Uncorrected
CellBender
B.
Sequencing saturation (%)
C.
Mean reads per nucleus (×1000)
D.
Mitochondrial %
Uncorrected
CellBender
E.
Doublet rate (%)
scDblFinder
Vireo (homotypic-corr.)
03-1A
03-1B
03-2
04
05-duck
05
06
07
08
PG3
PG4A
PG4B
F.
Mitochondrial %
G.
Ribosomal %
H.
Phase (%)
G1
S
G2M
g00 (3,525)
g06 (2,509)
g07 (3,450)
g02 (3,949)
g04 (2,053)
g11 (4,945)
g12 (2,786)
g03 (4,271)
g05 (2,543)
g08 (5,406)
g09 (3,299)
g01 (4,112)
g10 (2,401)
g17 (7,049)
g22 (2,618)
g23 (4,685)
g24 (6,162)
g15 (5,107)
g19 (3,508)
g16 (2,637)
g25 (2,268)
g32 (1,590)
g35 (2,665)
g13 (2,457)
g14 (2,906)
g18 (1,975)
g21 (2,290)
17-Md-13 (6,539)
g26a (3,649)
g26b (3,472)
g28 (3,410)
g34 (3,112)
g36 (8,363)
g37 (7,456)
g39 (6,944)
g38 (7,106)
g40 (5,220)
g43 (2,780)
g42 (2,820)
g41 (2,917)
g49 (1,507)
17-KC-1 (1,706)
25-8 (2,398)
g46 (1,977)
g51 (1,274)
g52 (1,194)
g50 (1,271)
g45 (2,056)
g48 (1,620)
g47 (1,921)
g44 (2,302)
g53 (753)
25-5..7
25-11..14
21-5..8
17-9..12
13-5..8
13-11..14
17-A..D
Md+KC (GEX03)
duck (GEX04)
I.
Variance ratio
Cumulative variance
30 PCs
Principal Component
J.
-0.050
-0.057
Uncorrected
Harmony
Silhouette
(lower = better)
K.
Overall: 0.197
Global res=0.2
RBC (n=80)
Cardiomyocyte (n=269)
Notochord (n=407)
Ventral neural prog. (n=701)
Dorsal neural prog. (n=907)
Diff. neuron (n=923)
Otic placode (n=948)
Endothelial (n=1,241)
Pharyngeal endoderm (n=2,190)
Epithelial (n=2,276)
Dermomyotome (n=2,622)
Neural crest (n=3,179)
Sclerotome (n=4,639)
Splanchnic mesenchyme (n=5,961)
Neural progenitor (n=10,666)
LPM mesenchyme (n=12,991)
Mean silhouette score


L.
% W-chromosome UMIs
Male (n=29)
Female (n=23)
M.
% Z-chromosome UMIs
g00 (3,525)
g06 (2,509)
g07 (3,450)
g02 (3,949)
g04 (2,053)
g11 (4,945)
g12 (2,786)
g03 (4,271)
g05 (2,543)
g08 (5,406)
g09 (3,299)
g01 (4,112)
g10 (2,401)
g17 (7,049)
g22 (2,618)
g23 (4,685)
g24 (6,162)
g15 (5,107)
g19 (3,508)
g16 (2,637)
g25 (2,268)
g32 (1,590)
g35 (2,665)
g13 (2,457)
g14 (2,906)
g18 (1,975)
g21 (2,290)
17-Md-13 (6,539)
g26a (3,649)
g26b (3,472)
g28 (3,410)
g34 (3,112)
g36 (8,363)
g37 (7,456)
g39 (6,944)
g38 (7,106)
g40 (5,220)
g43 (2,780)
g42 (2,820)
g41 (2,917)
g49 (1,507)
17-KC-1 (1,706)
25-8 (2,398)
g46 (1,977)
g51 (1,274)
g52 (1,194)
g50 (1,271)
g45 (2,056)
g48 (1,620)
g47 (1,921)
g44 (2,302)
g53 (753)
25-5..7
25-11..14
21-5..8
17-9..12
13-5..8
13-11..14
17-A..D
Md+KC (GEX03)
duck (GEX04)


N.
G6
G9
G3
G2
GT correlation (r)


O.
HH25 (8, 25,615)
HH21 (12, 29,759)
HH17 (6, 21,888)
HH17/16 (16, 71,365)
HH16 (2, 9,518)
HH13 (8, 18,788)
PC1 (sex-adj.)
Number of cells


P.
12 snRNA-seq libraries (cellranger count)
CellBender
ambient RNA correction
cellSNP-lite
variant genotyping
scDblFinder
doublet scoring
vireo genotype demultiplexing (per sequencing pool)
Main pool (7 libraries)
Universal SNPs (665K)
GEX05 + GEX06
25-series L/R
GEX07 + GEX08
13-series L/R
HH13 clutch-mate recovery
PG3 + GEX07
13-5..8
PG4A/B + GEX08
13-11..14
33 embryos
* 25-8 rescued via dp500 single-embryo recovery
GEX03
Pool-specific variants
(Mallard + KC breeds)
10 embryos
GEX04,
GEX0S-duck
De novo variants
(cellSNP mode 2b)
9 duck embryos
Cross-pool donor consolidation
Pseudobulk AF correlation I barcode concordance I suffix matching
Sex check
W/Z chromosome ratio
Unimodal per genotype
= single-embryo purity
Laterality check
L/R half assignment
Asymmetric genes confirm
correct side labeling
AF cross-validation
Cross-pool identity
Same embryo yields
r > 0.80 across pools
52 / 53 embryos confirmed
QC filters
genes [200, 8000] I UMIs >= 500 I mito <= 5% I vireo singlets only
181,159 vireo singlets I 12 libraries I 52 embryos
176,933-cell clustered atlas downstream (Figs. 4, S7)


Q.
Khaki Campbell (n=4)
Mallard (n=8)
Pearson r (genotype profile)
g36
g38
g41
17-KC-1
g39
g37
17-Md-13 (06)
17-Md-13 (12)
g43
g49
g40
g42